

DATE	
EXPT	7.1 Power Spectral Density Estimation and Spectral Analysis of Noisy Sinusoidal Signals Periodogram Objectives: Method in MATLAB

Program:

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Parameters
```

```
fs = 1000; % Sampling Frequency (Hz)
```

```
t = (0:999)/fs; % Time Vector
```

```
%% Signal Generation
```

```
s = sin(2*pi*100*t) + 0.5*sin(2*pi*200*t); % Original Signal
```

```
n = 0.3*randn(size(t)); % Gaussian Noise
```

```
x = s + n; % Noisy Signal
```

```
%% Figure 1: Signal with Noise
```

```
figure;
```

```
plot(t, x, 'b', 'LineWidth', 1.2);
```

```
grid on;
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
title('Signal with Noise');
```

```
%% Figure 2: Power Spectral Density (PSD)
```

```
figure;
```

```
periodogram(x, [], [], fs);
```

```
title('Power Spectral Density using Periodogram');
```

```
%% Figure 3: Magnitude Spectrum
```

```
X = abs(fft(x));
```

```
f = (0:length(X)-1) * fs / length(X);
```

```
figure;
```

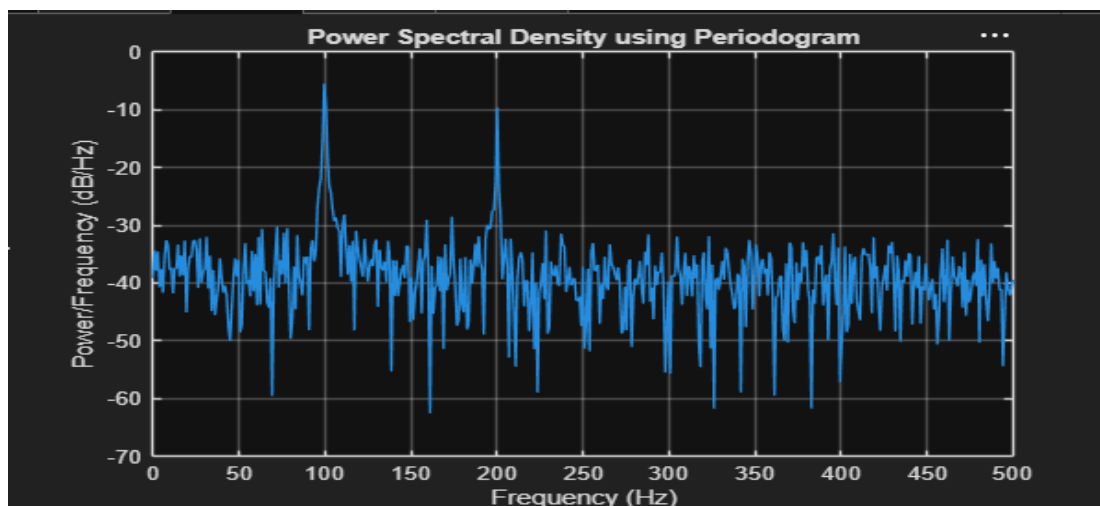
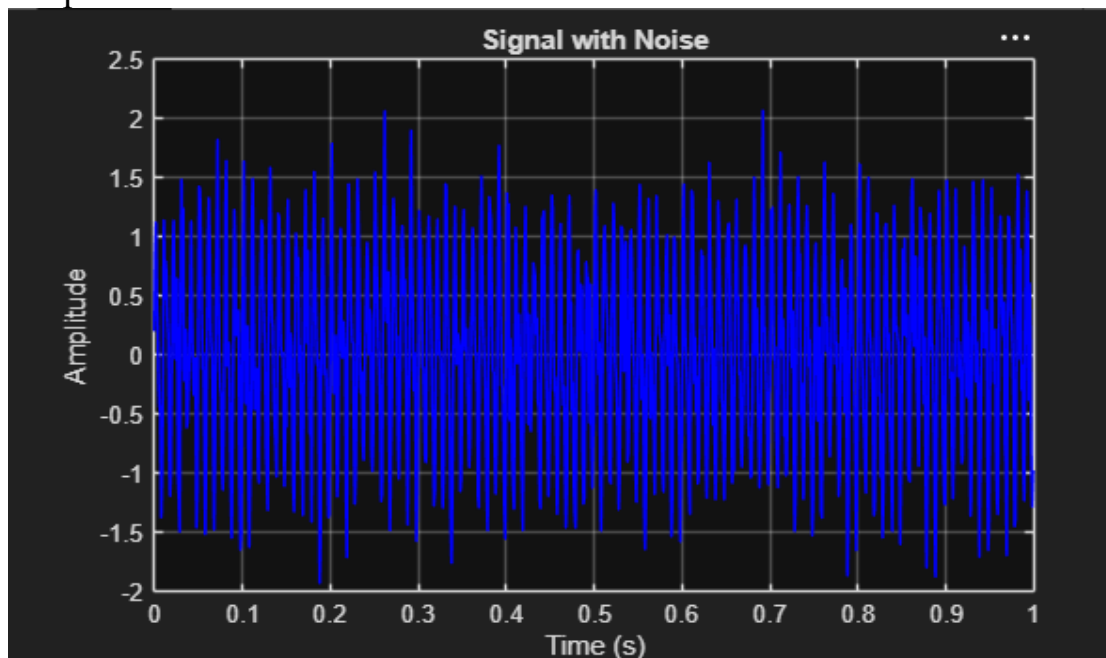
```
plot(f(1:length(X)/2), X(1:length(X)/2), 'r', 'LineWidth', 1.2);
```

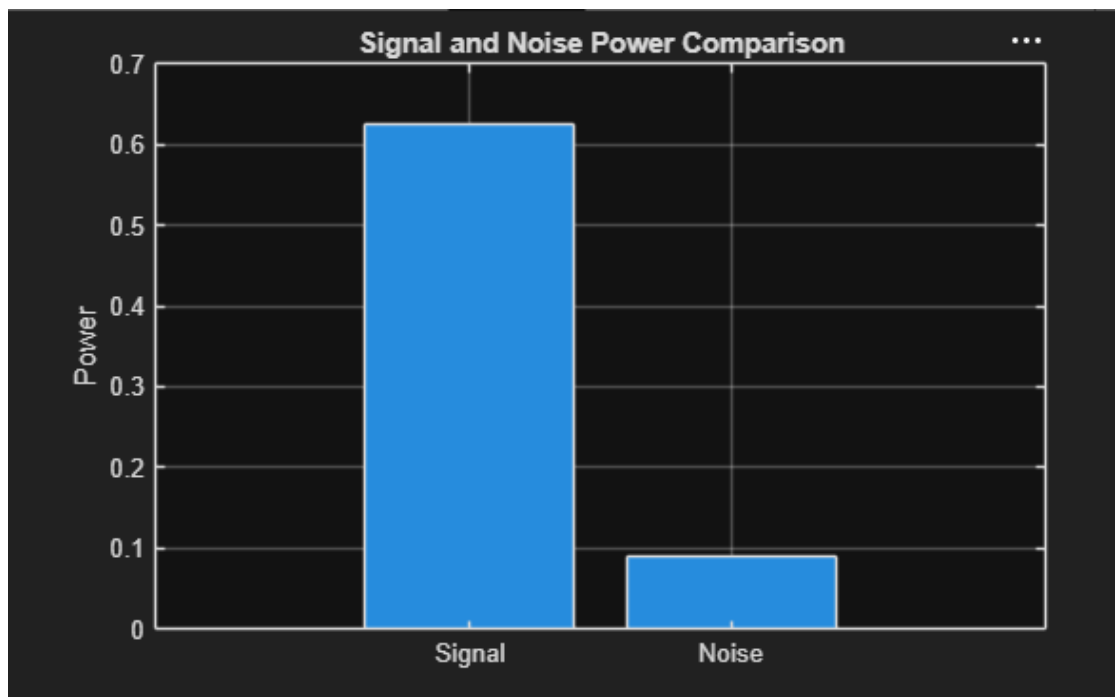
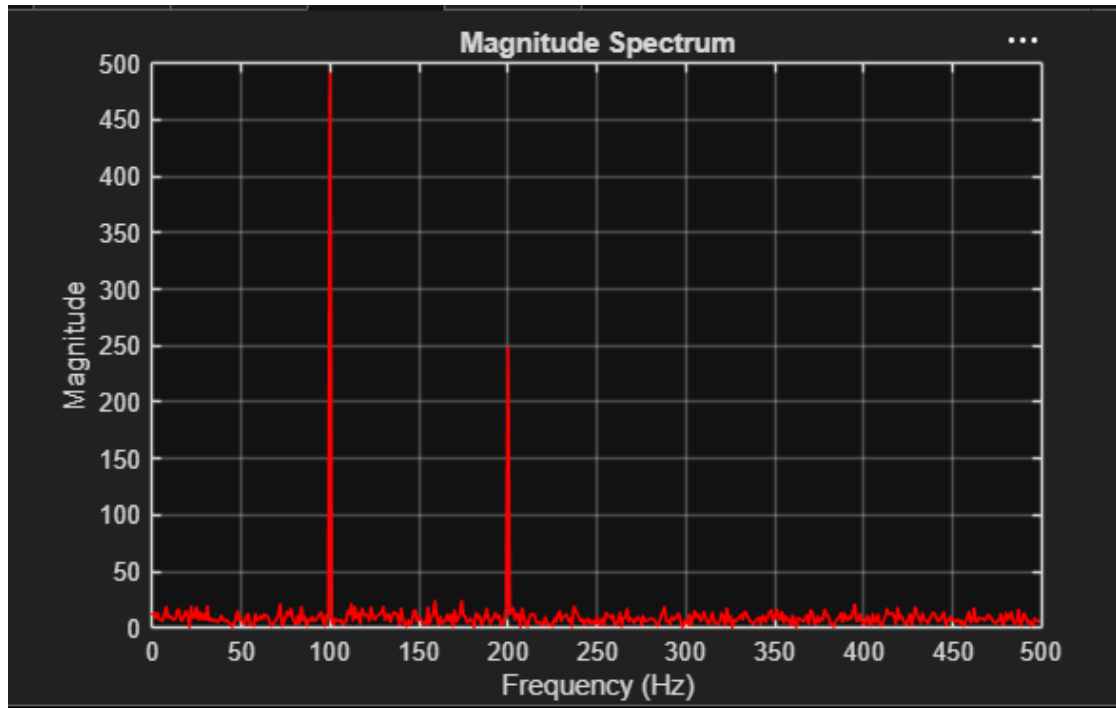
```
grid on;
```

```
noisePower = mean(n.^2);
```

```
figure;  
bar([signalPower noisePower]);  
set(gca, 'XTickLabel', {'Signal','Noise'});  
ylabel('Power');  
title('Signal and Noise Power Comparison');  
grid on;  
  
%% Display Results  
fprintf('Signal Power = %.4f\n', signalPower);  
fprintf('Noise Power = %.4f\n', noisePower);  
fprintf('SNR = %.2f dB\n', 10*log10(signalPower/noisePower));
```

Output:





DATE	
EXPT NO	7.2 Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, Hanning Windows in MATLAB

PROGRAM:

```

close all;

%% Signal Parameters
fs = 1000; % Sampling Frequency (Hz)
N = 256; % Number of Samples t =
(0:N-1)/fs; % Time Vector

% Signal (Two Close Frequencies)
x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);

%% FFT Parameters NFFT
= 2048;
f = (0:NFFT/2-1)*(fs/NFFT);

%% Window Functions
w = {rectwin(N), hamming(N), hann(N)};
names = {'Rectangular','Hamming','Hanning'};

P = cell(1,3); pks =
cell(1,3); loc =
cell(1,3);

%% Windowing and FFT
for i = 1:3
% Apply Window
xw = x(:) .* w{i};

% FFT
X = fft(xw, NFFT);

% Normalize Magnitude Spectrum P{i} =
abs(X(1:NFFT/2));
P{i} = P{i}/max(P{i}); end

%% Figure 1: Individual Spectra figure;

for i = 1:3 subplot(3,1,i);
plot(f,20*log10(P{i}),'k','LineWidth',1.5); xlabel('Frequency
(Hz)'); ylabel('Magnitude (dB)');
title(['Spectrum using ' names{i} ' Window']); xlim([0 250]);
grid on; end

%% Figure 2: Window Comparison figure;
plot(f,20*log10(P{1}),'k','LineWidth',1.5); hold on;
plot(f,20*log10(P{2}),'r','LineWidth',1.5);
plot(f,20*log10(P{3}),'b','LineWidth',1.5);

legend(names); xlabel('Frequency (Hz)');

```

```

ylabel('Magnitude (dB)'); title('Window
Comparison'); xlim([50 170]);
grid on;

%% Figure 3: Window Shapes figure;

for i = 1:3 subplot(3,1,i);
plot(w{i},'LineWidth',1.5);
xlabel('Samples');
ylabel('Amplitude');
title([names{i} ' Window']); grid on;
end

%% Peak Detection for i =
1:3

[pks{i}, loc{i}] = findpeaks(P{i}, f, ...
'MinPeakHeight',0.3,... 'MinPeakDistance',5);

fprintf('\n%s Window Detected Frequencies:\n', names{i}); disp(loc{i});

end

%% Figure 4: Peak Detection (Hamming Window) figure;
plot(f,20*log10(P{2}),'b','LineWidth',1.5); hold on;
plot(loc{2},20*log10(pks{2}),...
'ro','MarkerFaceColor','r');

xlabel('Frequency (Hz)'); ylabel('Magnitude
(dB)');
title('Detected Frequencies (Hamming Window)'); xlim([50 170]);
grid on;

%% Display Main Spectral Peaks fprintf('\nMain
Spectral Peaks:\n');

for i = 1:3
[~, idx] = max(P{i});
fprintf('%s Window Peak = %.2f Hz\n', names{i}, f(idx)); end

```

Output:

